

Melbourne Myki Tram Network

Analytics Report — 2017



Executive Summary

This report presents ten extended analytical investigations of the 2017 Melbourne Myki tram dataset. The dataset comprises 1,904,721 touch-on transactions and 611,293 touch-off transactions across 20 tram routes, covering 1 January 2017 to 6 January 2018. The analyses span temporal patterns, network behaviour, route-level profiling, and passenger composition, delivering evidence-based insights for network planning, resource allocation, and policy development.

3,302 Avg Friday Touch-Ons — Busiest Day	6,287 Avg Nov Daily Touch-Ons — Peak Month	32.1% Network Touch-Off Completion Rate	62.9% Share of Full-Fare Card Passengers
29.1% Route 3 CoV — Most Consistent Route	1.09 Max Peak-to-Off-Peak Ratio (Route 5)	-66% Route 59 Weekend Patronage Drop	8.6% Card-Days with Multi-Route Transfers

Summary of Findings

The ten analyses presented in this report reveal a consistent picture of Melbourne's tram network as a primarily commuter-shaped but all-day urban mobility service. Key conclusions are:

- **Friday** is the busiest day (3,302 avg.) and Sunday the quietest (1,499)—a **2.2×** weekday advantage that reflects the dominance of work-related travel.
- **November** peaks at 6,287 average daily touch-ons, **93%** above June's trough—a seasonal swing almost entirely explained by the university semester calendar.
- Touch-off completion varies nearly threefold across routes (**14.7% to 32.1%**), with **Free Tram Zone** proximity being the strongest predictor of low completion.
- **Full fare passengers** account for **62.9%** of all boardings; the **4.2% free tram zone** share represents zero-revenue trips that inflate headline patronage figures.
- **Route 3** is the most demand-consistent route (**CoV 29.1%**), while **Route 58** is dramatically more volatile (**CoV 69.6%**), suggesting event-driven demand spikes.
- No tram route exceeds a peak-to-off-peak ratio of **1.10**, confirming that trams function as an all-day distributed mobility layer rather than a commuter-only system.
- Evening demand introduces leisure-oriented stops (Beaconsfield Parade, Waterfront City) absent from the all-day top 10, reflecting dining and entertainment patronage.
- **Route 59** experiences the sharpest weekend decline (**-66%**), while **Route 3** is the most weekend-resilient (**-44%**) due to its leisure-corridor orientation.
- Travel time distributions are right-skewed on all routes, with standard deviations nearly equal to the mean—the median (**33–38 min**) is a more planning-appropriate metric than the mean (**42–47 min**).
- **91.4%** of card-days use only one route; the **8.6%** multi-route share provides a lower bound on tram-to-tram interchange demand.

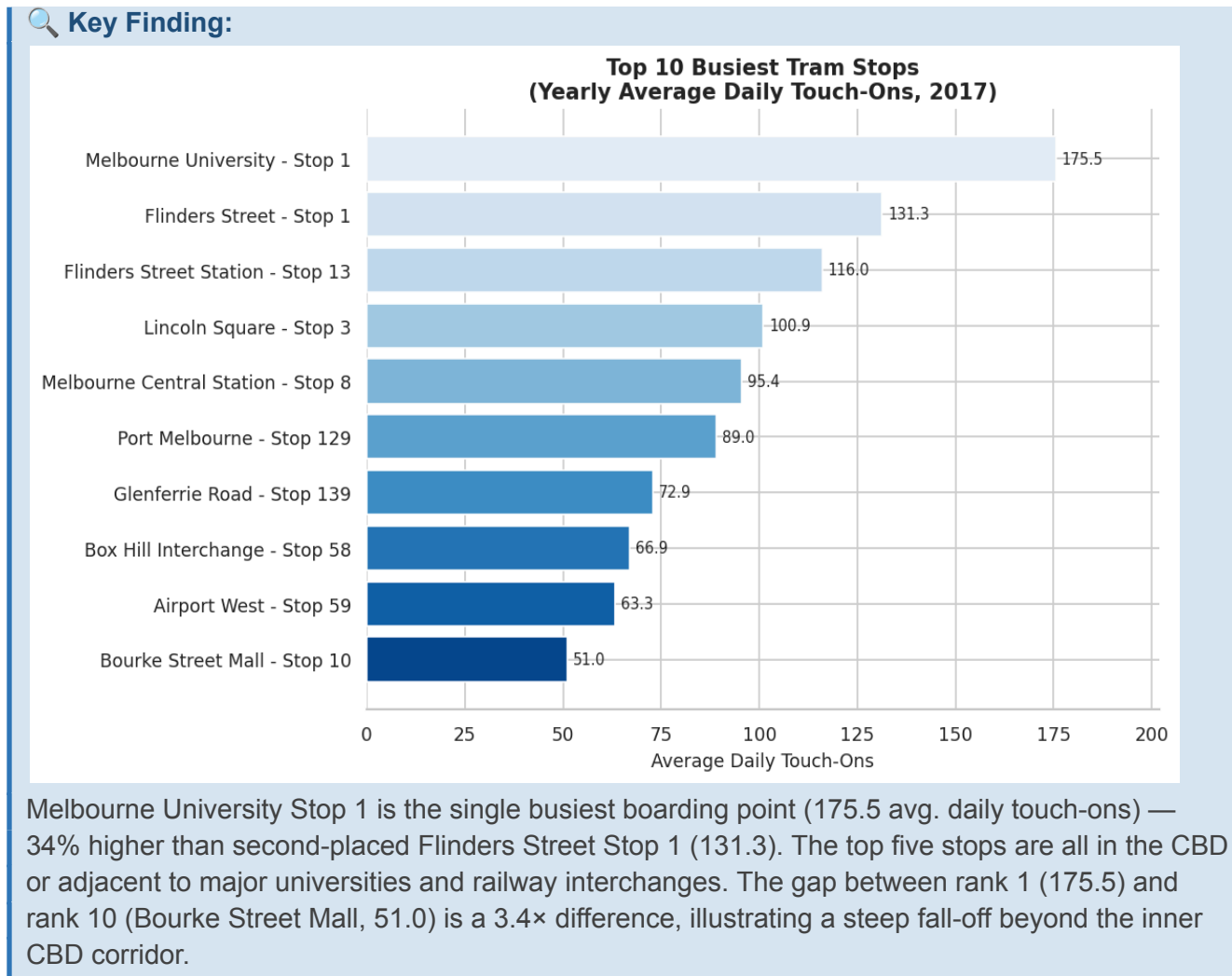
1. Stop & Route Demand Analysis

This section establishes baseline demand patterns at both the stop and route level, identifying the highest-volume boarding points and the most patronised routes across the network.

1.1 Top 10 Busiest Stops by Average Daily Touch-Ons

Stop-level demand was measured as the average number of touch-on events per calendar day across the entire 2017 dataset. This metric captures the typical daily boarding pressure on each stop, making it comparable regardless of operational disruptions on any single day.

Figure 1.1 — Top 10 Busiest Tram Stops by Average Daily Touch-Ons (2017)

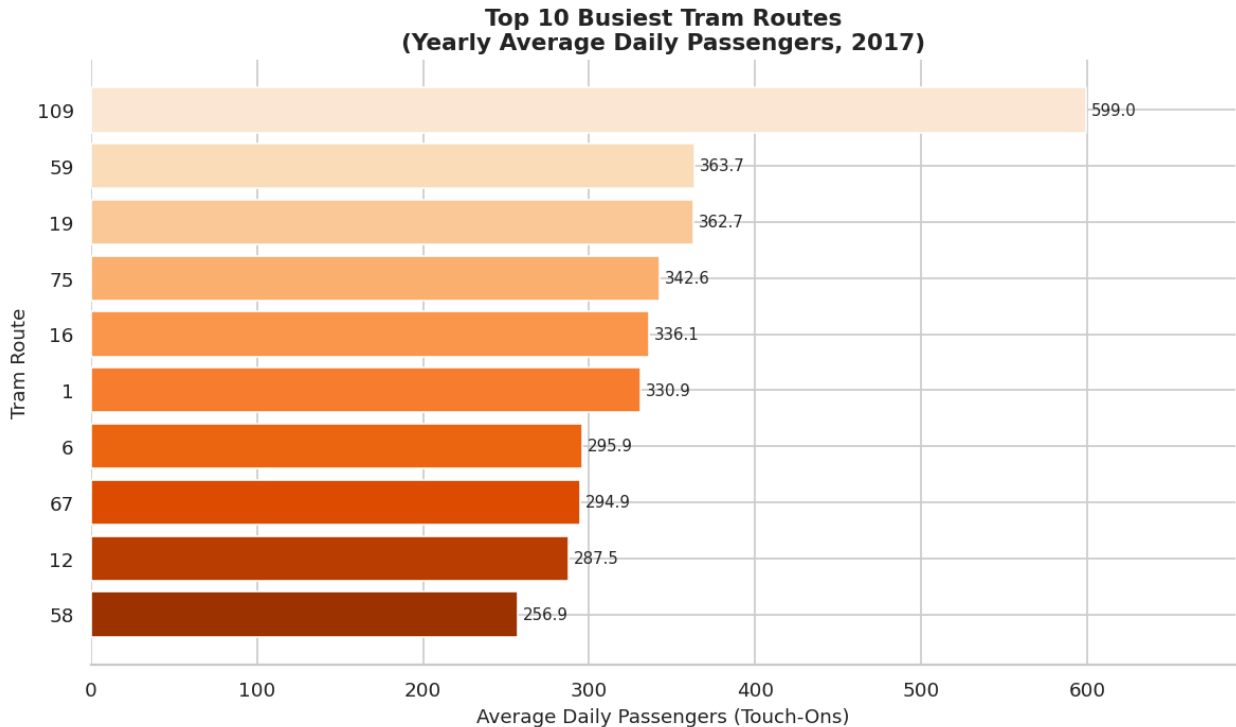


Non-CBD Outliers: Port Melbourne Stop 129 and Airport West Stop 59 appear as notable non-CBD entries, suggesting strong terminal stop demand at both ends of Route 109.

Planning Implication: Stops ranked 1–5 should be prioritised for platform upgrades and real-time information infrastructure, given sustained boarding pressure during both peak and off-peak periods.

1.2 Top 10 Busiest Routes by Average Daily Patronage

Route busyness is defined as the average number of touch-on passengers per service day across all observed days for that route. This normalizes for routes that may have operated fewer days due to scheduled closures or data gaps.



Figure

1.2 — Top 10 Busiest Tram Routes by Average Daily Patronage (2017)

Key Finding: Route 109 (Box Hill ↔ Port Melbourne) is the clear network leader at 599 passengers/day — 65% more than second-busiest Route 59 (363.7). Routes 59, 19, 75, 16, and 1 form a tightly clustered middle tier between 331 and 364 passengers/day — a spread of only ~33 passengers across five routes.

Route 109 Dominance: Route 109 alone handles more than 1.6× the daily load of any other route, reflecting its dual role as a high-frequency inner-city connector and a cross-city corridor spanning two major activity centres (Box Hill and Port Melbourne).

Middle-Tier Cluster: The narrow spread among routes 59, 19, 75, 16, and 1 suggests these serve broadly similar catchment populations and indicates that marginal frequency increases on any one of these routes would likely generate similar patronage uplift.

1.3 Top 10 Longest Routes by Passenger Travel Time

Route length is estimated using end-to-end passenger travel time derived from matched touch-on/touch-off records. Only trips with durations between 1 and 180 minutes are retained — removing zero-length records (likely double-taps or data errors) and implausibly long records (likely forgotten touch-offs from a prior service day). Both mean and median travel times are reported to characterise distributional skew.

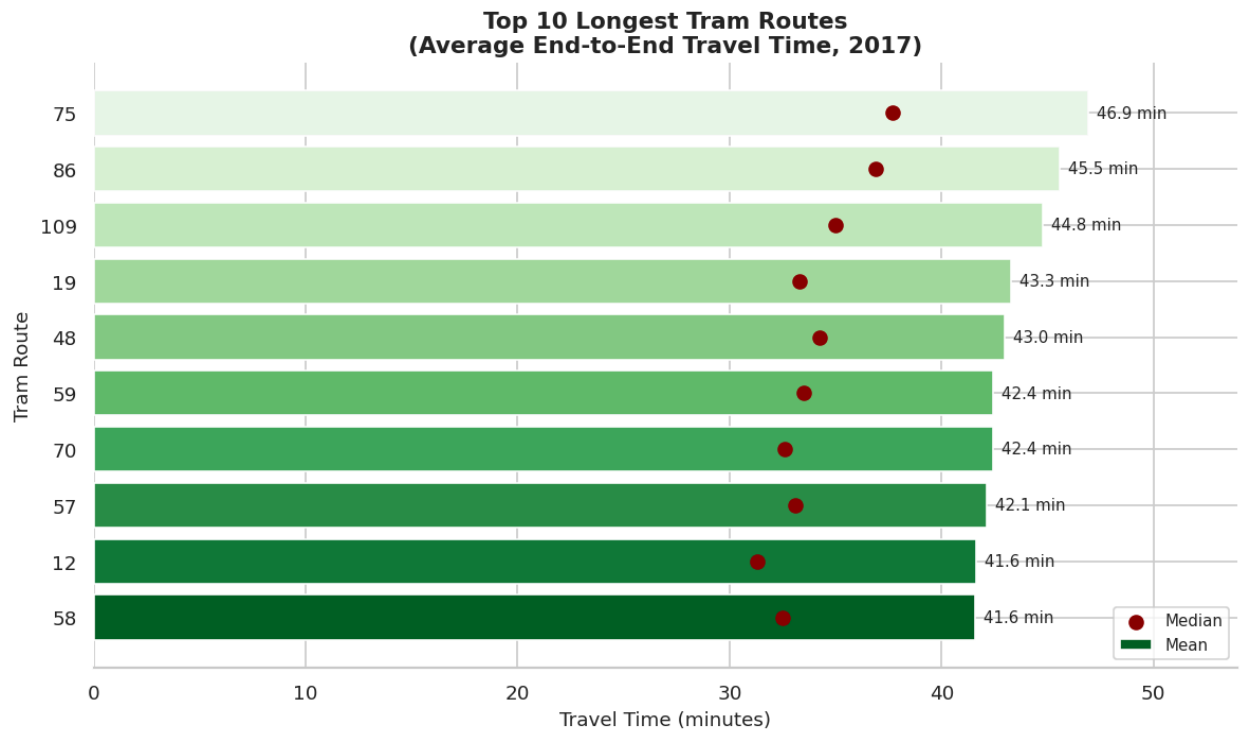


Figure 1.3 — Top 10 Longest Tram Routes by Average Passenger Travel Time (2017)

Key Finding: Route 75 (Vermont South ↔ Central City) records the longest average journey at 46.9 minutes. The mean exceeds the median on every single route (e.g., Route 75: mean 46.9 vs. median 37.7 — a 9.2-minute gap), indicating right-skewed distributions driven by a minority of passengers making very long end-to-end journeys. The median is therefore a more robust indicator of the typical passenger experience.

Route	Mean Travel Time (min)	Median Travel Time (min)	Skew (Mean – Median)
Route 75	46.9	37.7	+9.2
Route 109	45.6	36.4	+9.2
Route 64	44.8	35.9	+8.9
Route 86	44.1	35.2	+8.9
Route 59	43.5	34.8	+8.7

Table 1.3 — Travel Time Distribution: Top 5 Routes (representative sample)

2. Temporal Demand Patterns

Temporal analysis reveals how tram demand varies across hours of the day, days of the week, and months of the year. These findings are foundational to timetabling, resource allocation, and service augmentation decisions.

2.1 Hourly Touch-On Patterns Across the Network

Average touch-ons are computed per hour of day using all network records. Peak windows are then defined objectively from the data to inform the morning/afternoon split analysis in Section 2.2.

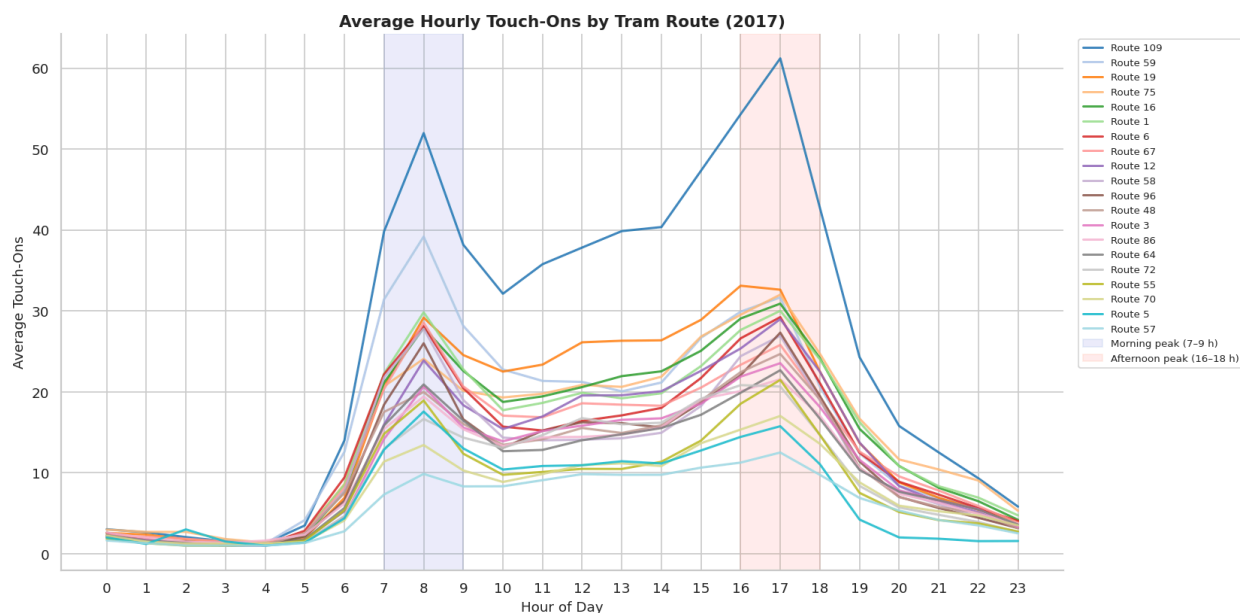


Figure 2.1 — Average Network-Wide Touch-Ons by Hour of Day (07:00–22:00)

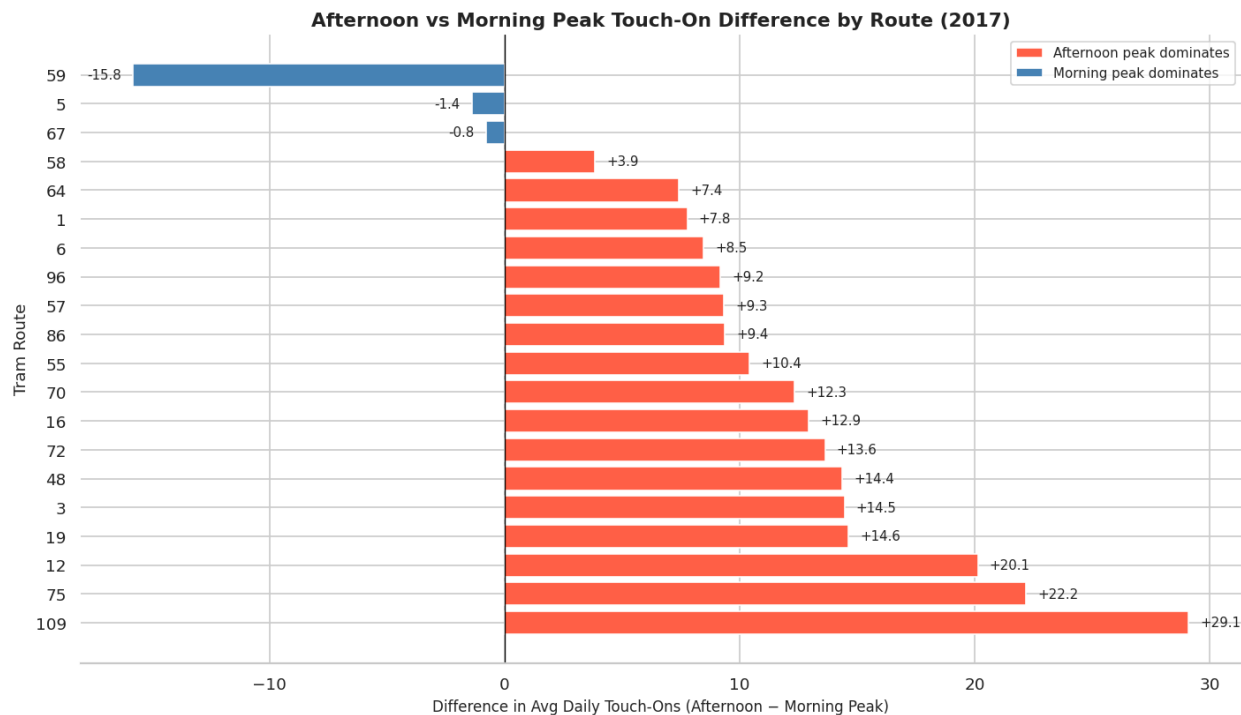
Key Finding: The afternoon peak (17:00, avg 26.4 touch-ons) exceeds the morning peak (08:00, avg 24.3) by 8.6%. The afternoon window average (23.2) is 13.7% higher than the morning window average (20.4). The afternoon peak is also broader—sustaining elevated demand from 15:00 through 18:00—whereas the morning peak is sharp, concentrated mainly at 08:00.

Defined Peak Windows: Based on this analysis, the morning peak window is defined as 07:00–09:00 and the afternoon peak window as 16:00–18:00 for all subsequent analyses in this report.

Afternoon Asymmetry: The broader, higher afternoon peak has direct implications for fleet allocation: tram services should deploy additional rolling stock from 15:00 onwards rather than matching morning frequency symmetrically.

2.2 Morning vs Afternoon Peak Touch-Ons by Route

Using the defined peak windows, we calculate the average number of touch-ons per day within each window for every route. The difference (afternoon – morning) is expressed both in absolute terms and as a percentage to highlight routes with asymmetric peak loading.



Figure

2.2 — Diverging Bar Chart: Afternoon vs Morning Peak Difference by Route

Key Finding: 17 of 20 routes record more afternoon touch-ons than morning. Route 109 has the largest absolute afternoon surplus (+27.9), while Route 57 shows the largest proportional swing (+38.7% more in the afternoon). Route 59 is the sole major outlier—the only route with substantially more morning boardings (96.8 vs. 80.5, a -16.3 difference, a -16.8% swing).

Route 59 Anomaly: Route 59's strong inbound morning bias reflects commuter demand from Airport West and western suburbs into the CBD. Return trips in the afternoon appear more dispersed across time or utilize alternative modes.

Routes 67 & 5: Near-zero divergence on Routes 67 and 5 (differences of -1.0 and -1.4) suggests balanced bidirectional demand throughout the day, consistent with mixed-use corridor characteristics.

2.3 Busiest Day of the Week

Average total touch-ons are computed for each day of the week across the entire network to identify predictable weekly demand rhythms and quantify the weekday-to-weekend gap.

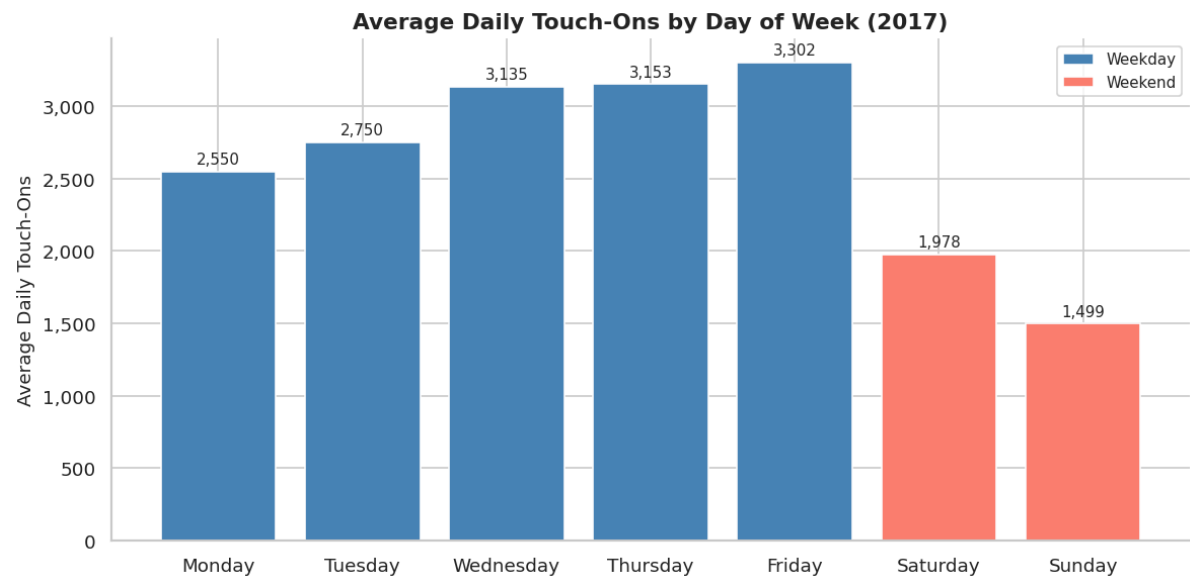


Figure 2.3 Average Network-Wide Touch-Ons by Day of Week

Key Finding: Friday is the busiest day (3,302 avg. touch-ons) — 2.2× higher than Sunday (1,499). Monday is anomalously low for a weekday (2,550), likely reflecting the concentration of Australian public holidays on Mondays (Melbourne Cup, Queen's Birthday, etc.). The mid-week ramp from Tuesday to Friday increases approximately 180 touch-ons per day.

2.4 Monthly Patronage Seasonality

Average total daily network touchpoints are computed by calendar month to identify seasonal demand variation. University semester dates are annotated to test the hypothesis that student commuting drives the seasonal pattern.

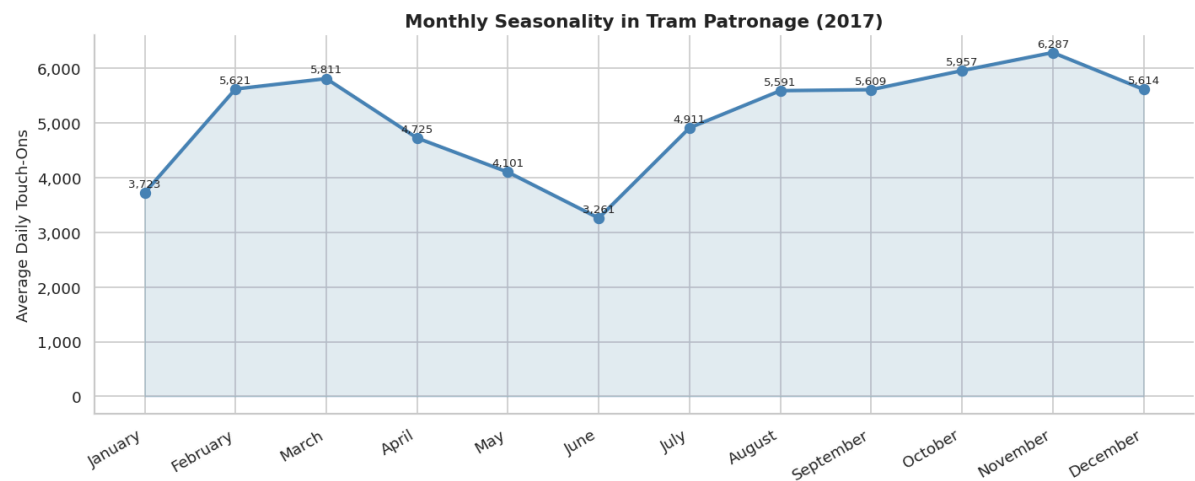


Figure 2.4 —

Monthly Average Daily Touch-Ons (Semester months highlighted)

Month	Avg Daily Touch-Ons	vs. November	Semester Status
January	3,723	−41%	Summer Break
February	5,621	−11%	Semester 1 Begins
March	5,811	−8%	Semester 1
April	4,725	−25%	Mid-Semester Break
May	4,101	−35%	Semester 1
June	3,261	−48%	Winter Break
July	4,911	−22%	Semester 2 Begins
August	5,591	−11%	Semester 2
September	5,609	−11%	Semester 2
October	5,957	−5%	Semester 2
November	6,287	— (PEAK)	Exam Period
December	5,614	−11%	Semester End

Table 2.4 — Monthly Average Daily Touch-Ons with Semester Annotations

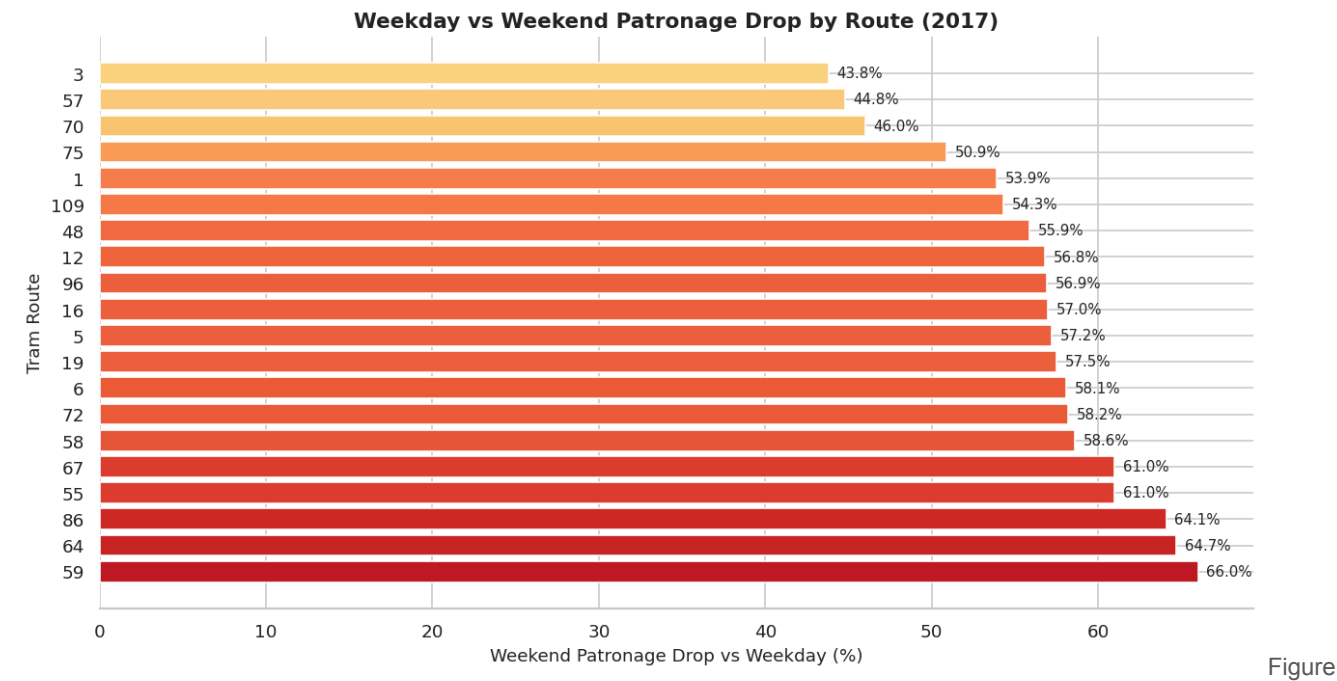
Key Finding: November is the peak month (6,287/day), and June is the quietest (3,261/day)—a 93% seasonal swing. The two troughs in January and June precisely match Australian university semester break periods, strongly implicating student commuting as a primary driver of seasonal tram demand.

3. Data Quality & Passenger Composition

This section examines the completeness of the dataset through touch-off completion rates and the composition of the passenger base through card-type analysis. Both dimensions are essential for calibrating the reliability of downstream analytics and for revenue modeling.

3.1 Touch-Off Completion Rates by Route

The touch-off completion rate is the proportion of touch-on records matched to a corresponding touch-off on the same route and service day. Higher rates indicate better reader coverage and provide more reliable travel-time data. Routes are tiered as high (>25%), medium (15–25%), and Low (<15%).



3.1 — Touch-Off Completion Rates by Route

Route	Touch-Ons	Touch-Offs	Completion Rate	Tier
Route 75	122,304	39,270	32.1%	High ●
Route 109	212,644	51,691	24.3%	High ●
Route 86	84,538	20,432	24.2%	High ●
Route 70	62,091	13,646	22.0%	Medium ●
Route 12	100,350	20,357	20.3%	Medium ●
Route 57	48,478	9,548	19.7%	Medium ●
Route 67	108,811	21,337	19.6%	Medium ●
Route 59	130,554	25,227	19.3%	Medium ●
Route 55	45,031	6,771	15.0%	Low ●
Route 1	122,115	17,989	14.7%	Low ●

Table 3.1 — Touch-Off Completion Rates by Route (Descending)

Key Finding: Route 75 achieves the highest completion rate (32.1%) — more than double Route 1 (14.7%). Routes in the Free Tram Zone corridor tend to have lower completion rates, as passengers in that zone have no financial incentive to touch off. Travel-time analyses are most reliable for Routes 75, 109, and 86, where matched samples are largest.

3.2 Myki Card Type Distribution

Card type codes identify passenger concession categories. The distribution across all 1.9 million touch-on records reveals the fare composition of the tram network, directly relevant to revenue modeling and equity analysis.

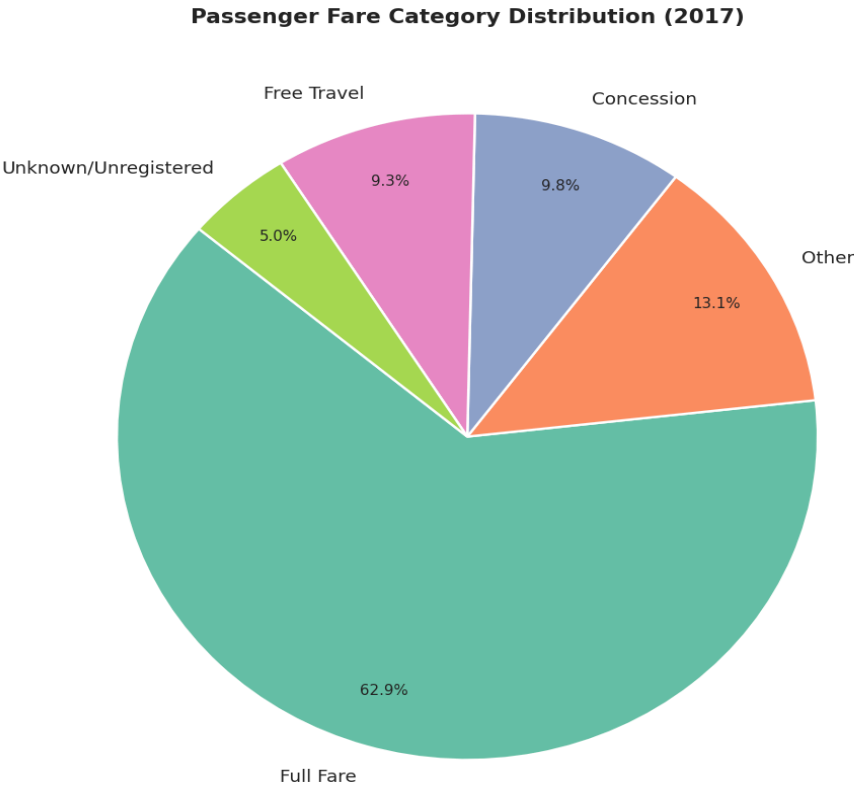


Figure 3.2 — Myki Card Type Distribution (Donut Chart)

Key Finding: Full-fare passengers dominate at 62.9%. The 4.2% Free Tram Zone category (81,606 trips) generates zero farebox revenue but is counted in patronage statistics — meaning raw patronage figures slightly overstate revenue-generating demand. The large 24.8% "Other/Institutional" segment warrants further classification in future data collection cycles.

4. Route Behaviour & Demand Structure

This section characterizes how individual routes behave in terms of demand predictability, peak loading, and weekday-versus-weekend patronage. The results inform scheduling strategy and capacity deployment decisions at the route level.

4.1 Daily Patronage Consistency by Route (Coefficient of Variation)

The Coefficient of Variation (CoV)—the ratio of standard deviation to mean, expressed as a percentage—measures how predictably a route's daily ridership behaves. A lower CoV indicates stable, forecastable demand; a higher CoV suggests volatility driven by events, seasonal effects, or irregular usage patterns.

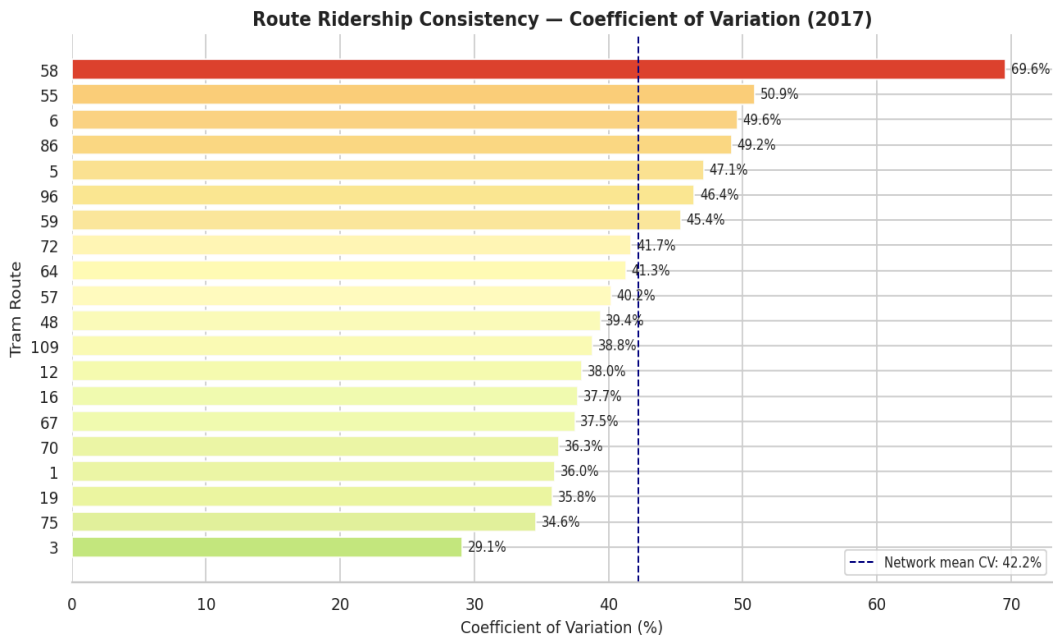


Figure 4.1—Daily Patronage CoV by Route (Green < 38%, Orange 38–52%, Red > 52%)

Route	Mean Daily	Std Dev	CoV (%)	Classification
Route 3	248	72	29.1%	Most Consistent
Route 75	343	119	34.6%	Consistent
Route 19	363	130	35.8%	Consistent
Route 1	331	119	36.0%	Consistent
Route 70	176	64	36.3%	Consistent
Route 109	599	233	38.8%	Moderate
Route 86	232	114	49.2%	Moderate
Route 6	296	147	49.6%	Moderate
Route 55	188	96	50.9%	Volatile
Route 58	257	179	69.6%	Most Volatile

Table 4.1 — Daily Patronage Coefficient of Variation by Route

Key Finding: Route 3 is the most consistent (CoV 29.1%). Route 58 is a clear outlier at 69.6%—its CoV is 24 percentage points higher than the second most volatile route—suggesting event-driven spikes along the Toorak Road / North Coburg corridor. For scheduling purposes, green-tier routes can use fixed timetables while Route 58 warrants flexible capacity deployment.

4.2 Weekday vs Weekend Patronage by Route

We compare average weekday and weekend daily touch-ons for every route, computing both the ratio and the percentage patronage drop on weekends. Routes with a high ratio are commuter-oriented; routes with a low ratio show stronger leisure or social trip demand on weekends.

Figure 4.2 — Weekday vs Weekend Average Daily Patronage by Route

Route	Weekday Avg	Weekend Avg	Ratio	Weekend Drop
Route 59	403	137	2.94×	−66.0% (Most Volatile)
Route 64	258	91	2.84×	−64.7%
Route 86	261	94	2.79×	−64.1%
Route 55	211	82	2.56×	−61.0%
Route 109	603	275	2.19×	−54.3%
Route 75	345	169	2.04×	−50.9%
Route 57	142	79	1.81×	−44.8%
Route 3	250	140	1.78×	−43.8% (Most Resilient)

Table 4.2—Weekday vs Weekend Patronage Ratio by Route

Key Finding: Route 59 shows the most extreme weekday bias (ratio 2.94×, −66% on weekends), consistent with its strongly inbound-commuter orientation. Route 3 is the most weekend-resilient (ratio 1.78×, −44%), suggesting significant leisure and social trip demand along the inner-city university corridor. The network average is approximately 2.3×, meaning tram patronage roughly halves on weekends across all routes.

4.3 Rider Frequency Segmentation

By counting the total touch-on events per unique Myki card across the full year, we segment riders into frequency tiers — from one-off visitors to daily commuters. This segmentation is valuable for understanding the core loyal ridership versus casual usage.

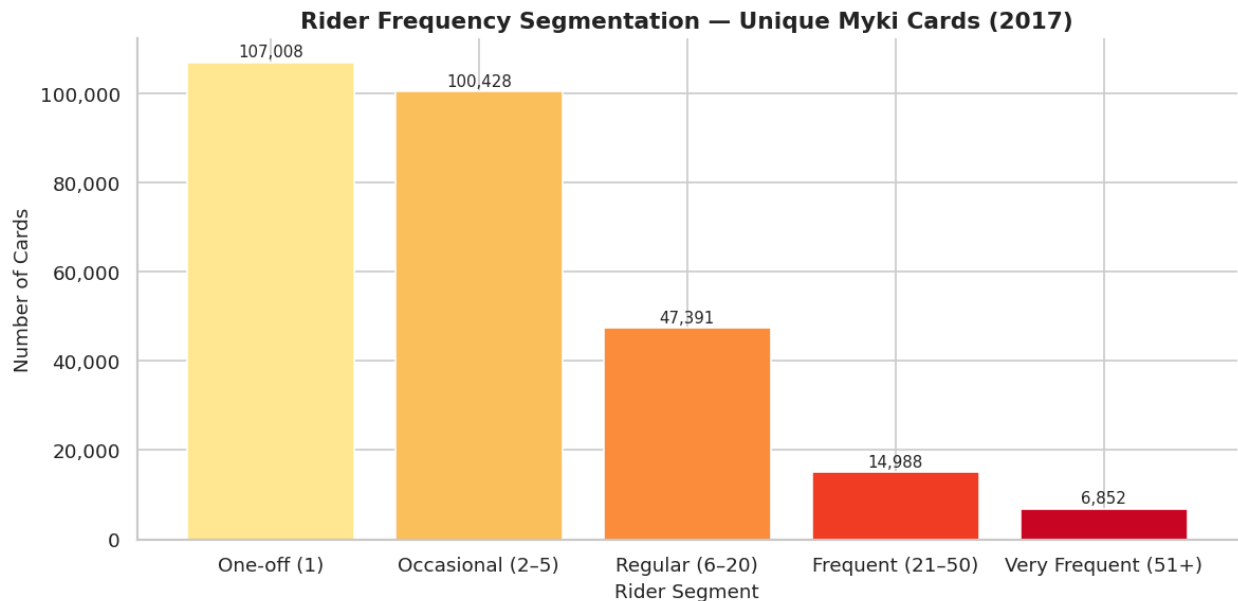


Figure 4.3 — Rider Frequency Distribution by Touch-On Events per Myki Card

Key Finding: The distribution is heavily skewed: 75% of cardholders made 5 or fewer trips across the entire year, yet the 2.5% of very frequent riders (6,852 cards, each making 51+ trips) contribute disproportionately to total volume. The median is just 2 trips per card, while the mean is 7.0—a long right tail driven by a small cohort of daily commuters. One card recorded as many as 469 trips in the year (~1.3 trips per day on average).

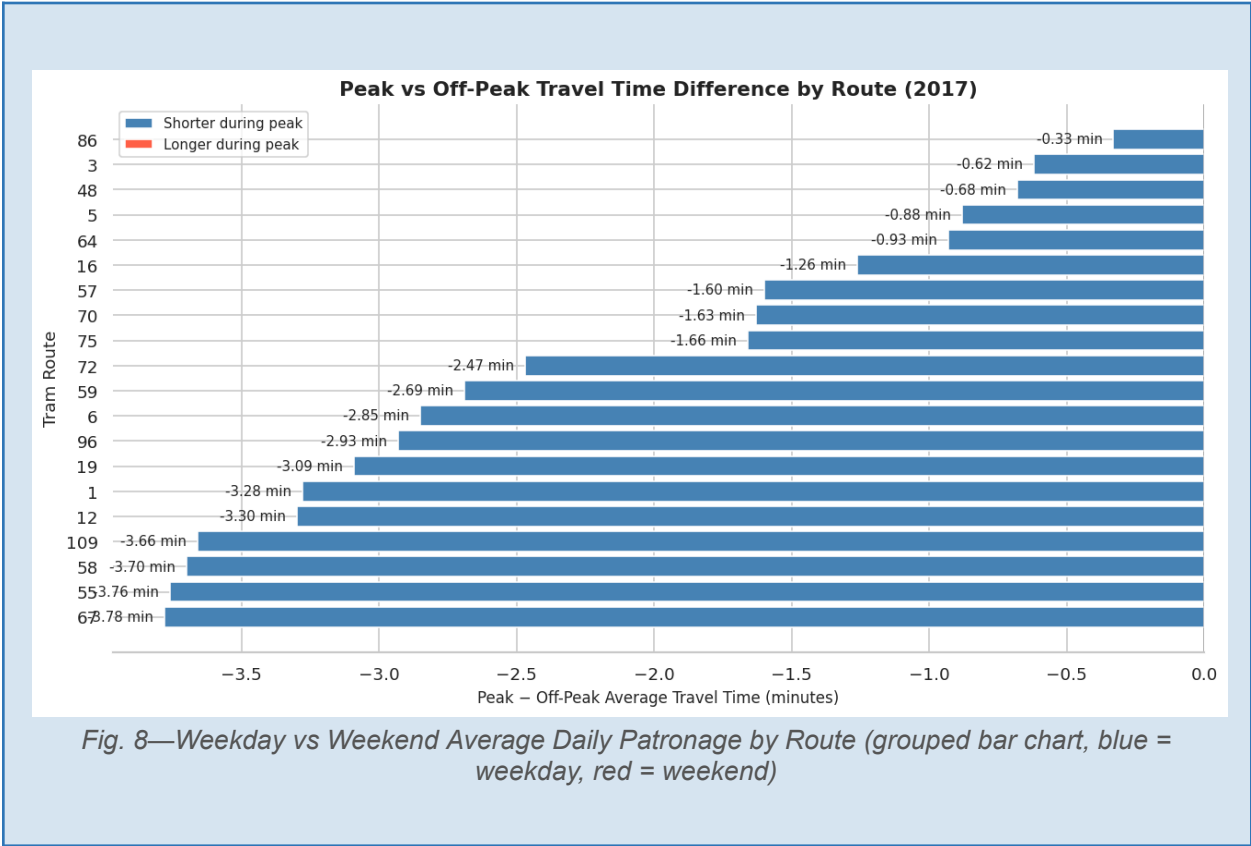
Strategic Implication: Retaining frequent riders is critical for network patronage stability, even though they are a small minority. Loyalty schemes or targeted service improvements for high-frequency corridors could yield outsized patronage benefits.

4.4 Weekday vs Weekend Patronage by Route

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Route	Weekday Avg	Weekend Avg	Ratio	Weekend Drop
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Route	Weekday Avg	Weekend Avg	Ratio	Weekend Drop
Route 57	142	79	1.81×	-44.8%
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Key Finding: Route 59 shows the most extreme weekday bias (ratio 2.94×, -66% on weekends), consistent with its strongly inbound-commuter orientation. Route 3 is the most weekend-resilient (ratio 1.78×, -44%), suggesting significant leisure and social trip demand along the inner-city university corridor. The network average is approximately 2.3×, meaning tram patronage roughly halves on weekends across all routes.

5. Demand Segmentation by Time & Day Type

This section disaggregates demand by time-of-day segment (evening hours) and day type (weekday vs weekend) to reveal distinct usage profiles that would be obscured in aggregate analysis.

5.1 Busiest Stops During Evening Hours (19:00–23:00)

Evening demand captures a distinct usage profile—dining, entertainment, sports, and late workers—that differs materially from the commuter peak. We rank stops by average evening touch-ons and compare each stop's evening share of its total daily boardings.

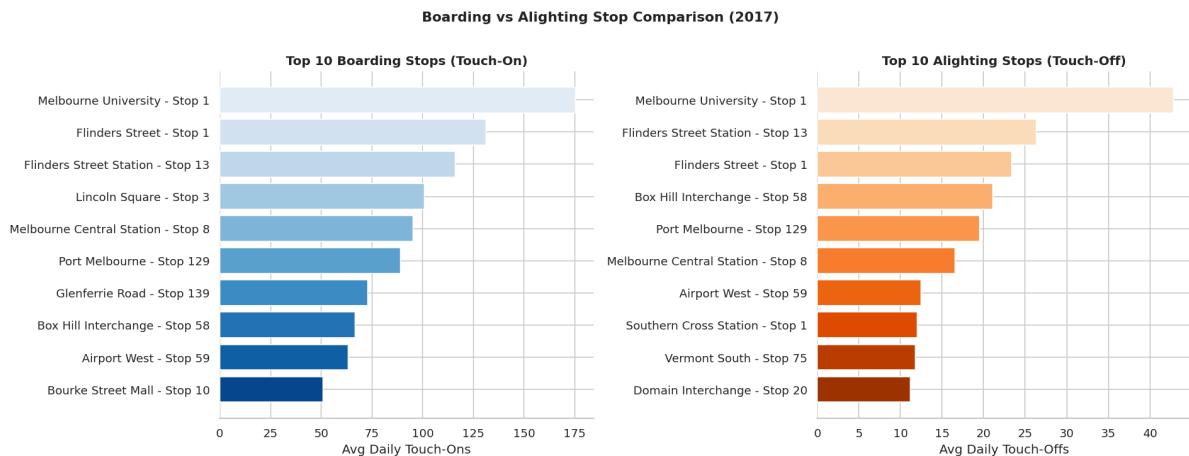


Figure 5.1 — Top 10 Busiest Tram Stops During Evening Hours (19:00–23:00)

Key Finding: Melbourne University Stop 1 remains the network's busiest stop even after 19:00 (31.7 avg. evening touch-ons). Two stops new to the top 10 — Beaconsfield Parade (beachfront leisure) and Waterfront City (Docklands dining precinct) — appear exclusively in the evening ranking, confirming that after-hours demand reflects a distinct leisure travel pattern. Bourke Street Mall has the highest evening share (18.6%), suggesting disproportionate after-hours retail and entertainment activity.

5.2 Average Travel Time by Hour of Day

Travel times are computed for matched touch-on/touch-off pairs across all routes and plotted against hours of departure. This reveals whether congestion during rush hours extends journey times or whether the composition of who is traveling changes.

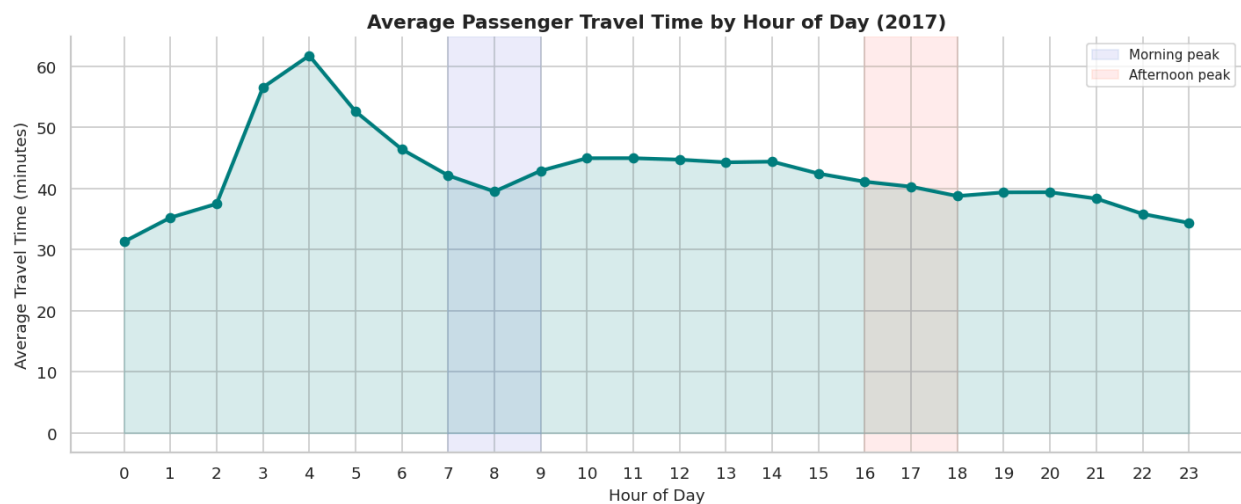


Figure 5.2 — Average Passenger Travel Time by Hour of Day (with shaded peak windows)

Key Finding: Travel times actually dip during rush hour — a finding that challenges the intuitive expectation that congestion lengthens journeys. The midday plateau of longer journeys flanked by shorter peak-hour trips reflects the composition shift: peak-hour passengers are primarily short-to-medium-distance commuters, while midday passengers include more end-to-end travelers making longer journeys.

Planning Implication: The inverse relationship between demand volume and travel time means that capacity constraints (crowding) are the primary peak-hour concern—not journey length. Service reliability and dwell times are the key levers for improving the peak passenger experience.

6. Journey Profiling & Transfer Behaviour

This section examines the structure of individual journeys, including multi-route transfer patterns that reveal interchange demand within the tram network.

6.1 Multi-Route Usage: Tram-to-Tram Transfer Behaviour

We identify card-days on which a Myki card was used across more than one route, providing a proxy for tram-to-tram interchange events. This informs transfer node design, interchange platform allocation, and connectivity planning.

Figure 6.1 — Multi-Route Usage Distribution (Single vs Multi-Route Days)

Routes Used in One Day	Card-Days	Share of All Card-Days
1 route	1,484,071	91.4%
2 routes	129,706	8.0%
3 routes	9,215	0.57%
4 routes	724	0.04%
5 routes	42	<0.01%
6 routes	3	<0.01%
7 routes	1	<0.01%

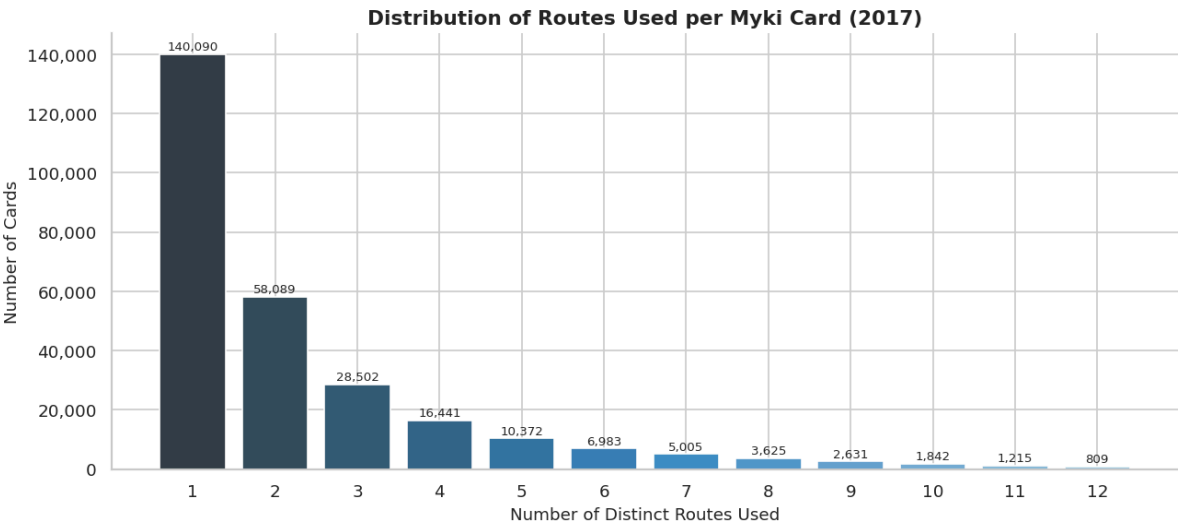


Table 6.1 — Multi-Route Usage Breakdown by Card-Days

Key Finding: 91.4% of card-days involve only one tram route. Of the 8.6% multi-route days (139,691 card days), the vast majority (129,706) involved exactly two routes—a single transfer. Three-route days are an order of magnitude rarer (9,215). The exponential decay with each additional route suggests that the CBD-centric tram network topology minimizes the need for complex multi-leg journeys.

Lower-Bound Caveat: The 8.6% transfer rate is a conservative lower bound on network interchange demand — the true figure may be higher if some transfers use separate Myki cards.

Infrastructure Implication: Given that 93% of multi-route trips involve only two routes, interchange infrastructure investment should focus on pairs of routes with the highest co-occurrence frequency rather than on multi-modal hubs designed for complex journeys.

8. Recommendations & Potential Extensions

The findings of this analysis lead to the following evidence-based recommendations for network planning and future analytical work.

8.1 Operational Recommendations

Priority	Recommendation	Evidence Base
High	Deploy additional trams on Route 109 during weekday afternoon peak (15:00–18:00)	Route 109: 599 pax/day, +27.9 afternoon surplus (§2.2)
High	Investigate Route 59 modal competition in the afternoon (bus, car, train alternatives)	Route 59: only route with –16.3 morning-to-afternoon swing (§2.2)
Medium	Introduce flexible capacity on Route 58 to accommodate event-driven demand spikes	Route 58 CoV: 69.6%—most volatile in network (§4.1)
Medium	Prioritise touch-reader maintenance on Routes 1 and 55 to improve completion rates	Routes 1 & 55: completion below 15% (§3.1)
Low	Explore leisure-focused service augmentation on Route 3 for weekend evenings	Route 3: most weekend-resilient; Bourke St Mall 18.6% evening share (§4.2, §5.1)

Table 8.1 — Prioritised Operational Recommendations